

Predicting Electricity Power Consumption for Tetuoan city in Zone 1

Final Project Report

STAT 385

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Team Members and Contributions

Team member	Contribution
Mahi Patel	Presentation Slide, Final report and doing analysis for zone 2 and zone 3(future works)
Rahee Chauhan	EDA, Data cleaning
Mustafa khan	Model building (linear regression, LASSO regression, decision tree)
Usman Tahir	Model evaluation, comparison

INTRODUCTION

Electricity demand and supply management is one of the most important problems in modern energy systems. Accurately predicting power output generated within a specific zone helps grid operators plan resources, reduce waste and avoid blackouts. In this project we focus on zone 1 power output using a dataset that contains weather related and time-based features that are recorded across multiple zones in power distribution network. We chose to focus on zone 1 specifically because it was the most complete and informative zone in the dataset. We also thought that analyzing one zone carefully gives us cleaner and more interpretable results than trying to predict all zones simultaneously. Future work could extend the models to other zones.

Research questions

- Can we predict power output in Zone 1 using weather and environmental features?
- Which variables (like temperature, humidity, wind speed or diffuse flows) are most strongly related to Zone 1 power output?
- Which statistical or machine learning model provides the best prediction performance for this task?
- Do nonlinear models like decision trees outperform simpler models like MLR for this dataset?

To answer these questions, we tried applying three different modeling approaches: Multiple Linear Regression, LASSO Regression and Decision Tree. We evaluated all models on the same training and testing splits to make fair comparison.

EXPLORATORY DATA ANALYSIS (EDA)

Before building any models, we explored the dataset carefully to understand its structure, identify patterns and decide which variables to include as predictors. This step helped us with modeling decisions.

Dataset Overview

The dataset contains data collected every 10 mins. It consists of energy related and weather variables from multiple zones. Each row represents one hour of data. Key variables include:

- Temperature(C): Ambient temperature in the zone
- Humidity (%): Relative humidity
- Wind Speed (km/h): Wind speed at the time of measurement
- General Diffuse Flows (GDF): Solar Diffuse irradiance
- Diffuse Flows (DF): Direct Diffuse irradiance
- Date and Time: Timestamp of the observation
- Zone 1 Power Consumption (MW): Target variable

We performed data cleaning by removing missing values, checking for outliers, and verifying that variable types are correct. The data-time variable extracted hour of day, day of week and month. These were added as additional predictors to capture time-based patterns in power usage.

Summary statistics

The table below shows summary statistics for main variables. Zone 1 power output ranged from roughly 18k to 52k megawatts with a mean near 34k megawatts. This indicates meaningful variation that our models need to capture

Variable	Min	Mean	Median	Max	Std Dev
Zone 1 Power (MW)	18,000	34,200	33,800	52,100	6,450
Temperature(C)	6.1	18.4	18.2	39.9	7.8
Humidity (%)	11.0	60.5	63.0	100.0	20.1
Wind Speed(km/h)	0.0	14.2	13.0	8.0	9.3
General Diffuse Flows	0.0	0.084	0.024	0.780	0.130
Diffuse Flows	0.0	0.093	0.027	0.815	0.142

Visualizations and key observations

We created several plots to understand how each predictor relates to zone 1 power output. Key findings from the EDA are as follows:

Temperature vs Zone 1 power

A scatter plot of temperature vs Zone 1 power showed nonlinear, U-shaped relationship. Power usage was high at both low temperatures i.e. during winters for heating and at high temperatures i.e. during summers for cooling. It was lower at moderate temperatures. This suggests that temperature is an important predictor, but it may benefit from a squared or polynomial term

Humidity vs Zone 1 power

Humidity showed moderate negative correlation with the power output. Higher humidity was associated with lower power consumption on average. The relationship was a bit noisy. This suggests that humidity alone can be a strong predictor

Diffuse Flows vs Zone 1 power

Both general Diffuse flows showed clear positive association with power consumption. Higher solar radiation periods tended to correspond with higher power demand. This is possibly because these measurements proxy for daytime activity levels.

Time Based Patterns

Hourly patterns showed clear peak in the morning around 8-9 Am and evening around 7-8PM. It was consistent with typical residential and commercial electricity usage. Day of week also showed differences, where weekdays had higher demand than weekends. These patterns confirmed that hour, day and month should be included as predictors.

Correlation Analysis

Correlation matrix showed that Diffuse Flow and General Diffuse Flows were strongly correlated with each other, which suggested multicollinearity. This motivated us to use LASSO regression, since LASSO can automatically shrink or eliminate redundant predictors. Temperature had moderate positive correlation with power, while humidity showed a negative correlation.

METHODS AND MODELS

We applied three modeling approaches to predict Zone 1 power output. For each method, we describe the model specification, variable selection, tuning, results and interpretation. All models used same training set (80% of data) and evaluated on the same testing set (20% of data). A random seed of set (2026) was used. The full set of predictors used across models includes:

Temperature	Humidity
Wind Speed	General Diffuse Flow
Diffuse Flows	Hour of Day
Day of week	Month

Method 1: Multiple Linear Regression (MLR)

Model specification

Multiple Linear Regression models the outcome as a weighted linear combination of predictors:

$$\text{Zone1_Power} = \beta_0 + \beta_1(\text{Temperature}) + \beta_2(\text{Humidity}) + \beta_3(\text{WindSpeed}) + \beta_4(\text{GDF}) + \beta_5(\text{DF}) + \beta_6(\text{Hour}) + \beta_7(\text{DayOfWeek}) + \beta_8(\text{Month}) + \varepsilon$$

Here ε is the random error term which is assumed to follow a normal distribution with mean 0.

Variable Selection and Model Fitting

We first fit a full model with all available predictors. We then examined p-value and variance inflation factors to check multicollinearity. General Diffuse Flows and Diffuse Flows had a variance inflation factor greater than 10. This confirms multicollinearity. We kept both in model since LASSO handles these issues in Method 2. Here we wanted to see the full linear model for comparison

Results

Results produced by Multiple Linear Regression model are as follows:

Predictor	Estimate	Std Error	T-value	p-value
Intercept	21,450	312	68.7	<0.001
Temperature	185.3	12.4	14.9	<0.001
Humidity	-42.8	5.7	-7.5	<0.001
Wind Speed	18.2	6.1	2.98	0.003
General Diffuse Flows	12,340	890	13.9	<0.001
Diffuse Flows	8,120	760	10.7	<0.001
Hour	210.5	18.3	11.5	<0.001
Day of Week	-85.4	22.1	-3.86	<0.001
Month	42.1	9.8	4.29	<0.001

The model achieved an adjusted R^2 of 0.72. This means that 72% of the variation in Zone 1 power is explained by predictors and all predictors were statistically significant at the 0.05 level.

Interpretation

Each 1°C increase in temperature results with an increase of about 185 MW in Zone 1 power, holding other predictors constant. Higher humidity is associated with lower power use. The positive coefficient for Hour suggests power usage increase through the day on average. The General Diffuse Flows and Diffuse Flows coefficient reflect strong daytime activity captures in solar radiation data

Conclusion and Limitation

Linear regression provides solid interpretable baseline. It assumes a linear relationship between predictors and outcomes. This may not fully capture the U-shaped temperature effect or other nonlinear patterns. The test RMSE was approximately 4,210 MW.

Method 2: Least Absolute Shrinkage & Selection Operator (LASSO) Regression

Model Specification

LASSO regression adds penalty on size of the coefficients to standard linear regression loss. The function is:

$$\text{Minimize: } \sum (y_i - \hat{y}_i)^2 + \lambda \sum |\beta_j|$$

Here lambda is a tuning parameter, and it controls how much shrinkage is applied. When lambda is large, coefficient shrinks towards 0 or are exactly set to 0.

Tuning Parameter Selection

We selected best value of lambda using 10-fold cross validation. The `cv.glmnet()` function in R was used to fit LASSO across the grid of lambda values and compute cross validation mean squared error (CV-MSE) for each. We used `lambda.min` as our final model. This is the lambda value that minimizes CV-MSE. The optimal lambda computed was approximately 0.012.

Variable Selection Results

After applying LASSO with optimal lambda, all 8 predictors were retained in the model. The coefficients for Diffuse Flows and General Diffuse Flows were slightly shrunk toward each other. This confirms that all predictors carry meaningful information, but LASSO helped reduce the instability caused by their high correlation.

Results and Interpretations

LASSO model achieved a test RMSE of approximately 4050 MW, which is slightly better than linear regression. The coefficient estimates were similar in direction to LOS estimates but with slightly smaller magnitudes due to shrinkage. This confirms the robustness of findings from Method 1 while providing some insurance against overfitting.

Conclusion

LASSO regression outperformed standard linear regression on the test set. It also provides automatic variable selection, which is useful when the number of predictors is large. In our case all predictors survived, which suggests that they are all useful.

Method 3: Decision Tree

Model Specification

A decision tree is a nonlinear model that recursively splits the data on predictor values to create homogeneous groups. At each split, the algorithm selects the predictor and a split point that most reduces the residual sum of squares. The result is a tree structure where each leaf node predicts the mean outcome value of observations that fall in the group.

Tuning Parameter Selection: Pruning with cp

An unpruned tree can grow very deep and overfit the training data. We used cost complexity pruning to find the right tree size. The complexity parameter cp controls this. A larger cp forces a simpler/smaller tree, whereas small cp will allow more splits. We selected optimal cp using 10-fold cross validation with `rpart` package in R, choosing the cp value that minimized cross validation error. The optimal cp was 0.002, which results in a tree with approximately 14 terminal nodes. This value of cp represents a deliberate tradeoff. The tree is complex enough and captures the nonlinear patterns in data, but not so deep that it memorizes noise in the training set. A plot of cp vs. cross validated error. This confirmed that further reducing cp beyond 0.002 led to only marginal improvements in cross validation error while substantially increasing tree complexity. This made the pruned tree the preferred choice.

Results

The most important splits in the tree were based on Hour of Day that captured morning and evening demand peaks, General Diffuse Flows that captures solar activity and Temperature that captures seasonal effects. The decision tree achieved a test RMSE of approximately 3890 MW, which is better than both linear regression approaches.

Interpretation

We found the tree model highly interpretable. For example, during nighttime hours with low diffuse flows, the predicted power output is in the low range. During daytime hours with high diffuse flows and moderate to high temperatures, the model predicts peak power usage. These splits align well with real world intuition about electricity demand patterns.

Conclusion

The pruned decision tree captures nonlinear relationships better than linear models and achieved the lowest test RMSE among the three methods. The cp selection via cross validation was essential. Without pruning, the tree severely overfitted the training data and performed worse on the test set.

MODEL COMPARISON and RECOMMENDATION

All 3 models were evaluated on the same 20% test set using RMSE (root mean squared error) and R-squared as performed metrics. RMSE measures the average prediction error in the same units as Zone 1 power, so smaller is better. R-squared measures the proportion of variation explained by model.

Model	Test RMSE(MW)	Test R-squared	Interpretable?	Handles Nonlinearity?
Multiple Linear Regression	4,210	0.714	Yes	No
LASSO Regression	4,050	0.727	Yes	No
Decision Tree (Pruned)	3,890	0.745	Yes	Yes

Recommendation:

We would recommend the Pruned Decision Tree as the best model for this dataset based on the following reasons:

- It achieved the lowest test RMSE (3,890MW) and the highest test R squared (0.745) among all three models.
- It naturally handles nonlinear relationships such as U-shaped temperature effect and time of day demand peaks, which linear models cannot fully capture
- It remains interpretable. The tree structure can be visualized, and splits provide real world explanation for the predictors
- The pruning step ensured the model generalized well to test set rather than overfitting

LASSO regression is our second recommendation. If interpretability in terms of linear coefficients is required, LASSO regression can be used since it improves on standard OLS by reducing the impact of multicollinear predictors. Though standard linear regression was the simplest and most transparent, it was the weakest performer,

AREAS OF FUTURE IMPROVEMENT

Based on feedback from professor and our analysis, several improvements could be considered in future work:

- Methods such as Ridge Regression, Random Forest, Gradient Boosting and Best Subset Selection could be compared. Tree methods like Random Forest and Boosting typically outperform simple decision trees by averaging many trees and reducing variance.
- We included hour, day of week, and month, but we could try adding more time features like public holidays, daylight saving transitions to improve predictions
- The dataset contains power data for multiple zones. Including zone specific features or building separate models per Zone could reveal spatial patterns
- Adding a squared temperature term in linear models could better capture the U shaped demand patterns observed in EDA.

CONCLUSION

This project is set out to predict Zone 1 electricity power output using weather and time-based features. Through careful exploratory data analysis, we identify the most important predictors and uncovered key patterns such as time-of-day demand peaks and a nonlinear temperature effect. We applied three modeling approaches: Multiple Linear regression, LASSO Regression, and A Decision Tree and evaluated them on the same test data.

Our main conclusions are:

- The decision tree was the best performing model achieving a test RMSE of 3890 MW and R squared of 0.745. The pruning step was critical to prevent overfitting.
- LASSO Regression offered slight improvement over standard linear regression and helped address multicollinearity between diffuse flow variables.
- The most important predictors across all models were Hour of Day, General Diffuse Flows, Temperature and Diffuse Flows. These variables reflect the daily activity cycle and weather conditions that drive electricity demand.
- Time features like hour, day of week and month improved prediction accuracy, which suggests that future models should consider even richer temporal features like holiday indicators

The models we built practically provide useful predictions and interpretable insights. They could help grid operators anticipate Zone 1 demand under varying weather conditions and plan to supply resources accordingly. Future world should explore methods like Random Forest and Gradient Boosting, which are likely to capture the complex interactions between time, weather and demand even more effectively.